Item No		Unit	Quantity	Rate	Amount

SECTION NO. 1

BILL NO. 1
PRELIMINARIES

BUILDING AGREEMENT AND PRELIMINARIES

The **JBCC** Principal Building Agreement (Edition 6.2 - May 2018) prepared by the Joint Building Contracts Committee shall be the applicable building agreement, amended as hereinafter described

The **JBCC** Principal Building Agreement **contract data** form an integral part of this **agreement**

The **JBCC** General Preliminaries (May 2018) published by the Joint Building Contracts Committee for use with the **JBCC** Principal Building Agreement (Edition 6.2 - May 2018) shall be deemed to be incorporated in these **bills of quantities**, amended as hereinafter described

The **contractor** is deemed to have referred to the abovementioned documents for the full intent and meaning of each clause

The clauses in the abovementioned documents are hereinafter referred to by clause number and heading only

Where any item is not relevant to this **agreement** such item is marked N/A signifying "not applicable"

Where standard clauses or alternatives are not entirely applicable to this **agreement** such amendments, modifications, corrections or supplements as will apply are given under each relevant clause heading and such amendments, modifications, corrections or supplements shall take precedence notwithstanding anything to the contrary contained in the abovementioned documents

	Carried Forward			R	**Not Priced**

Brought Forward		R	**Not Priced**

PREAMBLES FOR TRADES

The General Preambles for Trades 2017 published by the Association of South African Quantity Surveyors shall be deemed to be incorporated in these **bills of quantities** and no claims arising from brevity of description of items fully described in the said General Preambles will be entertained

Supplementary preambles and/or specifications are incorporated in these **bills of quantities** to satisfy the requirements of this project. Such supplementary preambles and/or specifications shall take precedence over the provisions of the General Preambles

The **contractor's** prices for all items throughout these **bills of quantities** shall take account of and include where applicable for all of the obligations, requirements and specifications given in the General Preambles and in any supplementary preambles and/or specifications

STRUCTURE OF THIS PRELIMINARIES BILL

Section A : A recital of the headings of the individual clauses in the aforementioned **JBCC** Principal Building Agreement

Section B : A recital of the headings of the individual clauses in the aforementioned **JBCC** General Preliminaries

Section C : Any special clauses to meet the particular circumstances of the project

PRICING OF PRELIMINARIES

Should the **contractor** select Option A in the **contract data** for the adjustment of **preliminaries**, the amounts entered against the relevant items in these **preliminaries** are to be divided into one or more of the three categories provided namely fixed (F), value related (V) and time related (T)

SECTION A: PRINCIPAL BUILDING AGREEMENT

Carried Forward		R	**Not Priced**

	Brought Forward			R	**Not Priced**
	<u>Interpretation (A1-A7)</u>				
1	Clause 1.0 - Definitions and interpretation **Pricing of bills of quantities** The **contractor** is to allow opposite each item for all costs in connection therewith. All prices to include, unless otherwise stated, for all materials, fabrication, conveyance and delivery, unloading, storing, unpacking, hoisting, labour, setting, fitting and fixing in position, cutting and waste (except where to be measured in accordance with the standard system of measurement), patterns, models and templates, plant, temporary works, returning of packaging, duties, taxes (other than Value Added Tax), imposts, establishment charges, overheads, profit and all other obligations arising out of this **agreement**. Value Added Tax (VAT) is to be separately stated on the summary page of these **bills of quantities** Items left unpriced will be deemed to be covered in prices against other items throughout these **bills of quantities** and no claim for any extras arising out of the **contractor's** omission to price any item will be entertained Prices for all **construction equipment**, temporary works, services and other items shall include for the supply, maintenance, operating cost and subsequent removal and making good as necessary **Abbreviated descriptions** The items in these **bills of quantities** utilise abbreviated descriptions. It is the intention that the abbreviated descriptions be fully described when read with the applicable measuring system and the relevant preambles and/or specifications. However, should the full intent and meaning of any description not be clear, the **contractor** shall, before submission of his tender, call for a written directive from the **principal agent**, failing which it shall be assumed that the **contractor** has allowed in his pricing for materials and workmanship in terms of international best practice **Legal status of contractor**				
	Carried Forward			R	**Not Priced**

	Brought Forward			R	**Not Priced**
	If the **contractor** constitutes a joint venture, consortium or other unincorporated grouping of two or more persons then: 1. These persons are deemed to be jointly and severally liable to the **employer** for the performance of this **agreement** 2. These persons shall notify the **employer** of their leader who has assigned authority to bind the **contractor** and each of these persons 3. The **contractor** shall not alter its composition or legal status without the prior written consent of the **employer** F:..... V:..... T:.....		Item		2 424 228.00
2	Clause 2.0 - **Law**, regulations and **notices** F:..... V:..... T:.....		Item		**Not Priced**
3	Clause 3.0 - Offer and acceptance F:..... V:..... T:.....		Item		**Not Priced**
4	Clause 4.0 - Cession and assignment F:..... V:..... T:.....		Item		**Not Priced**
5	Clause 5.0 - Documents **Value Added Tax** Provision is made in the summary page of these **bills of quantities** for the inclusion of Value Added Tax (VAT) **Priced document as specification** Clause 5.4 is deemed to be deleted				
	Carried Forward			R	2 424 228.00

Brought Forward			R	2 424 228.00
The **principal agent** shall decide which portion of the **priced document** may be used as a specification of **materials and goods** or methods, if any **Electronic issue of drawings** All drawings for this project will be issued electronically and the **contractor** shall be deemed to have received such drawings on the date that such drawings have been dispatched electronically [5.6] F:..... V:..... T:.....		Item		**Not Priced**
Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
6	Clause 6.0 - **Employer's agents** 6.1. Development Manager: Department of Infrastructure Development 6.2. Agent Service: Architect HDM Group 6.3. Agent Service: Quantity Surveyor Vumah Management 6.4. Agent Service: Civil Engineer Tlou Integrated Tech 6.5. Agent Service: Structural Engineer: Tlou Integrated Tech 6.6. Agent Service: Electrical Engineer HDM Group 6.7. Agent Service: Mechanical, Wet services and Fire Engineer HDM Group Engineers 6.8. Agent Service: Health and Safety TBA 6.9. Agent Service: Environmental Health TBA **6.1 Delegated authority** The authority of the **principal agent** to issue **contract instructions** [17.1] and perform duties for specific aspects of the **works** is delegated to **agents** as follows [6.2]. This does not preclude the **principal agent** from issuing such **contract instructions**:				
	Carried Forward			R	2 424 228.00

	Brought Forward		R	2 424 228.00
6.2. Architect				
6.2.1 Duties [6.2] :				
The architect is responsible for the architectural design, functional design and quality inspection of the **works**				
6.2.2 **Contract instructions** [6.2; 17.1] :				
6.2.2.1 Rectification of discrepancies, errors in description or quantity or omission of items in the **agreement** other than in the **JBCC** Principal Building Agreement				
6.2.2.2 Alteration to design, standards or quantity of the **works** provided that such **contract instructions** shall not substantially change the scope of the **works**				
6.2.2.3 The **site** [13.0]				
6.2.2.4 Compliance with the **law**, regulations and bylaws [2.1]				
6.2.2.5 Provision and testing of samples of **materials and goods** and/or of finishes and assemblies of elements of the **works**				
6.2.2.6 Opening up of work for inspection, removal or re-execution [23.2.4; 26.4.2]				
6.2.2.7 Removal or re-execution of work				
6.2.2.8 Removal or substitution of any **materials and goods**				
6.2.2.9 Protection of the **works**				
6.2.2.10 Making good physical loss and repairing damage to the **works** [23.2.2]				
6.2.2.11 Rectification of **defects** [21.2]				
	Carried Forward		R	2 424 228.00

	Brought Forward		R	2 424 228.00
	6.2.2.12 A **list for practical completion** specifying outstanding or defective work to be rectified to achieve **practical completion**, a **list for completion** and a **list for final completion** specifying outstanding or defective work to be rectified to achieve **final completion**			
	6.2.2.13 Expenditure of **budgetary allowances**, **prime cost amounts** and **provisional sums**			
	6.2.2.14 Appointment of a **subcontractor** [14.0; 15.0]			
	6.2.2.15 Work by **direct contractors** [16.0]			
	6.2.2.16 On suspension or termination, protection of the **works**, removal of **construction equipment** and surplus **materials and goods** [29.0]			
	6.3. Quantity surveyor			
	6.3.1 Duties [6.2] :			
	The quantity surveyor is responsible for all measurements, valuations, financial assessments and all other quantity surveying and cost control functions of the **works**			
	6.3.2 **Contract instructions** [6.2; 17.1] :			
	6.4. Civil engineer			
	6.4.1 Duties [6.2] :			
	The civil engineer is responsible for all aspects of civil and structural engineering design and quality inspection of the **works**			
	6.4.2 **Contract instructions** [6.2; 17.1] :			
	6.4.2.1 Rectification of discrepancies, errors in description or quantity or omission of items in the **agreement** other than in the **JBCC** Principal Building Agreement			
	Carried Forward		R	2 424 228.00

	Brought Forward		R	2 424 228.00
	6.4.2.2 Alteration to design, standards or quantity of the **works** provided that such **contract instructions** shall not substantially change the scope of the **works**			
	6.4.2.3 The **site** [13.0]			
	6.4.2.4 Compliance with the **law**, regulations and by laws [2.1]			
	6.4.2.5 Provision and testing of samples of **materials and goods** and/or of finishes and assemblies of elements of the **works**			
	6.4.2.6 Opening up of work for inspection, removal or re-execution [23.2.4; 26.4.2]			
	6.4.2.7 Removal or re-execution of work			
	6.4.2.8 Removal or substitution of any **materials and goods**			
	6.4.2.9 Protection of the **works**			
	6.4.2.10 Making good physical loss and repairing damage to the **works** [23.2.2]			
	6.4.2.11 Rectification of **defects** [21.2]			
	6.4.2.12 A **list for practical completion** specifying outstanding or defective work to be rectified to achieve **practical completion**, a **list for completion** and a **list for final completion** specifying outstanding or defective work to be rectified to achieve **final completion**			
	6.4.2.13 Expenditure of **budgetary allowances**, **prime cost amounts** and **provisional sums**			
	6.5. <u>Structural engineer</u>			
	6.5.1 Duties [6.2] :			
	The structural engineer is responsible for all aspects of civil and structural engineering design and quality inspection of the **works**			
	Carried Forward		R	2 424 228.00

	Brought Forward		R	2 424 228.00
6.5.2	**Contract instructions** [6.2; 17.1] :			
6.5.2.1	Rectification of discrepancies, errors in description or quantity or omission of items in the **agreement** other than in the **JBCC** Principal Building Agreement			
6.5.2.2	Alteration to design, standards or quantity of the **works** provided that such **contract instructions** shall not substantially change the scope of the **works**			
6.5.2.3	The **site** [13.0]			
6.5.2.4	Compliance with the **law**, regulations and by laws [2.1]			
6.5.2.5	Provision and testing of samples of **materials and goods** and/or of finishes and assemblies of elements of the **works**			
6.5.2.6	Opening up of work for inspection, removal or re-execution [23.2.4; 26.4.2]			
6.5.2.7	Removal or re-execution of work			
6.5.2.8	Removal or substitution of any **materials and goods**			
6.5.2.9	Protection of the **works**			
6.5.2.10	Making good physical loss and repairing damage to the **works** [23.2.2]			
6.5.2.11	Rectification of **defects** [21.2]			
6.5.2.12	A **list for practical completion** specifying outstanding or defective work to be rectified to achieve **practical completion**, a **list for completion** and a **list for final completion** specifying outstanding or defective work to be rectified to achieve **final completion**			
6.5.2.13	Expenditure of **budgetary allowances**, **prime cost amounts** and **provisional sums**			
	Carried Forward		R	2 424 228.00

	Brought Forward		R	2 424 228.00
	6.6. Mechanical, wet services and fire engineer			
	6.6.1 Duties [6.2] :			
	The mechanical, wet services and fire engineer is responsible for all aspects of mechanical engineering design and quality inspection of the **works** and, where appointed by the **employer** for quantity surveying services in respect of the mechanical installations, for all measurements, valuations, financial assessments and all other quantity surveying and cost control functions			
	6.6.2 **Contract instructions** [6.2; 17.1] :			
	6.6.2.1 Rectification of discrepancies, errors in description or quantity or omission of items in the **agreement** other than in the **JBCC** Principal Building Agreement			
	6.6.2.2 Alteration to design, standards or quantity of the **works** provided that such **contract instructions** shall not substantially change the scope of the **works**			
	6.6.2.3 Compliance with the **law**, regulations and bylaws [2.1]			
	6.6.2.4 Provision and testing of samples of **materials and goods** and/or of finishes and assemblies of elements of the **works**			
	6.6.2.5 Opening up of work for inspection, removal or re-execution [23.2.4; 26.4.2]			
	6.6.2.6 Removal or re-execution of work			
	6.6.2.7 Removal or substitution of any **materials and goods**			
	6.6.2.8 Protection of the **works**			
	6.6.2.9 Making good physical loss and repairing damage to the **works** [23.2.2]			
	6.6.2.10 Rectification of **defects** [21.2]			
	Carried Forward		R	2 424 228.00

	Brought Forward		R	2 424 228.00
6.6.2.11	A **list for practical completion** specifying outstanding or defective work to be rectified to achieve **practical completion**, a **list for completion** and a **list for final completion** specifying outstanding or defective work to be rectified to achieve **final completion**			
6.6.2.12	Expenditure of **budgetary allowances**, **prime cost amounts** and **provisional sums**			
6.7.	Electrical engineer			
6.7.1	Duties [6.2] :			
	The electrical engineer is responsible for all aspects of electrical engineering design and quality inspection of the **works** and, where appointed by the **employer** for quantity surveying services in respect of the electrical installations, for all measurements, valuations, financial assessments and all other quantity surveying and cost control functions			
6.7.2	**Contract instructions** [6.2; 17.1] :			
6.7.2.1	Rectification of discrepancies, errors in description or quantity or omission of items in the **agreement** other than in the **JBCC** Principal Building Agreement			
6.7.2.2	Alteration to design, standards or quantity of the **works** provided that such **contract instructions** shall not substantially change the scope of the **works**			
6.7.2.3	Compliance with the **law**, regulations and bylaws [2.1]			
6.7.2.4	Provision and testing of samples of **materials and goods** and/or of finishes and assemblies of elements of the **works**			
6.7.2.5	Opening up of work for inspection, removal or re-execution [23.2.4; 26.4.2]			
6.7.2.6	Removal or re-execution of work			
	Carried Forward		R	2 424 228.00

	Brought Forward		R	2 424 228.00
	6.7.2.7 Removal or substitution of any **materials and goods**			
	6.7.2.8 Protection of the **works**			
	6.7.2.9 Making good physical loss and repairing damage to the **works** [23.2.2]			
	6.7.2.10 Rectification of **defects** [21.2]			
	6.7.2.11 A **list for practical completion** specifying outstanding or defective work to be rectified to achieve **practical completion**, a **list for completion** and a **list for final completion** specifying outstanding or defective work to be rectified to achieve **final completion**			
	6.7.2.12 Expenditure of **budgetary allowances**, **prime cost amounts** and **provisional sums**			
	6.8. Health and safety consultant			
	6.8.1 Duties [6.2] :			
	The health and safety consultant is responsible for all aspects of health and safety of the **works**. Without derogating from the generality thereof, the health and safety consultant will perform the following specific functions and duties in respect of the health and safety aspects of the **works**. He shall:			
	6.8.1.1 Act as the **employer's agent** in terms of the Construction Regulations issued in terms of the Occupational Health and Safety Act,1993 as amended			
	6.8.1.2 Prepare and update the health and safety specification for the **works**			
	6.8.1.3 Agree with the **contractor** the health and safety plan for the **works**			
	6.8.1.4 Carry out regular audits to ensure adherence to the safety plan and compliance with the act and regulations			
	Carried Forward		R	2 424 228.00

	Brought Forward			R	2 424 228.00
	6.8.1.5 Stop the execution of the **works** where the agreed specification or plan is not adhered to				
	6.9. Environmental Health consultant				
	6.9.1 Duties [6.2] :				
	The Environmental health consultant is responsible for all aspects of environmental health of the **works**. Without derogating from the generality thereof, the environmental health consultant will perform the following specific functions and duties in respect of the environment health aspects of the **works** and the construction site environmental plan				
	6.8.1.1 Act as the **employer's agent** in terms of the Construction Regulations issued in terms of the National Health Act, 2003				
	F:..... V:..... T:.....		Item		**Not Priced**
7	Clause 7.0 - Design responsibility				
	F:..... V:..... T:.....		Item		**Not Priced**
	Insurances and securities (A8-A11)				
8	Clause 8.0 - **Works** risk				
	F:..... V:..... T:.....		Item		**Not Priced**
9	Clause 9.0 - Indemnities				
	F:..... V:..... T:.....		Item		**Not Priced**
10	Clause 10.0 - Insurances				
	F:..... V:..... T:.....		Item		**Not Priced**
11	Clause 11.0 - **Securities**				
	Guarantee for payment				
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
	The **employer** shall not provide to the **contractor** a **guarantee for payment** in the amount of N/A .....Rand (R N/A.....) [11.5.1]. The **contractor** shall consequently waive his lien or right of continuing possession of the **works** [11.10]				
	Extension of waiver of lien				
	The **contractor** shall ensure that a waiver of lien is included in all subcontracts and that the **works** executed on the **site** are kept free of all liens and other encumbrances at all times [11.10]				
	F:..... V:..... T:.....		Item		**Not Priced**
	Execution (A12 - A17)				
12	Clause 12.0 - Obligations of the **parties**				
	Office accommodation				
	The **contractor** shall provide, maintain and remove on **practical completion** air conditioned office accommodation with suitable tables and chairs for meetings to be held on the **site**. Such offices shall be kept clean and fit for use at all times [12.2.18]				
	Notice board				
	The **contractor** shall erect in a position approved by the **principal agent**, maintain and remove on **practical completion** a notice board recommended by the South African Institute of Architects and as approved by the **principal agent** listing the names and logos of the **employer**, the **contractor** and the professional consultants. No subcontractor or supplier notice boards may be erected unless permission is granted by the **principal agent** for such notice boards to be erected [12.2.18]				
	Statutory and other notices				
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
	The **contractor** shall submit and/or comply with all statutory and other notices that may be required by any local or other authority in order not to cause any delay to the commencement of the **works** by the **contractor**. The **contractor** shall pay all deposits or fees in this regard It is, however, specifically recorded that the **employer** shall be responsible for the timeous approval of building plans by any local or other authorities and the payment of any fees or charges related thereto F:..... V:..... T:.....		Item		**Not Priced**
13	Clause 13.0 - Setting out F:..... V:..... T:.....		Item		**Not Priced**
14	Clause 14.0 - Nominated **subcontractors** F:..... V:..... T:.....		Item		**Not Priced**
15	Clause 15.0 - Selected **subcontractors** F:..... V:..... T:.....		Item		**Not Priced**
16	Clause 16.0 - **Direct contractors** **Attendance on direct contractors** In respect of **direct contractors** the **contractor** shall: 1. Designate an area for the **direct contractor** to establish a temporary office and workshop and storage of equipment and materials 2. Allow the use of personnel welfare facilities, where provided				
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
	3. Provide water, lighting and single phase electric power to a position within 50m of the place where the direct contract work is to be carried out, other than fuel or power for commissioning of any installation				
	4. Permit the **direct contractor** to use erected scaffolding, hoisting facilities, etc provided by the **contractor**, in common with others having the like right, while it remains erected on the **site** [16.1]				
	F:..... V:..... T:.....		Item		**Not Priced**
17	Clause 17.0 - **Contract instructions**				
	Site instructions				
	Instructions issued on **site** are to be recorded in a site instruction book which is to be supplied and maintained on **site** by the **contractor**				
	F:..... V:..... T:.....		Item		**Not Priced**
	<u>Completion (A18 - A24)</u>				
18	Clause 18.0 - Interim completion		N/A		
19	Clause 19.0 - **Practical completion**				
	F:..... V:..... T:.....		Item		**Not Priced**
20	Clause 20.0 - Completion in **sections**				
	F:..... V:..... T:.....		Item		**Not Priced**
21	Clause 21.0 - **Defects** liability period and **final completion**				
	F:..... V:..... T:.....		Item		**Not Priced**
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
22	Clause 22.0 - **Latent defects** liability period F:..... V:..... T:.....		Item		**Not Priced**
23	Clause 23.0 - Revision of the date for **practical completion** **Substitution of materials and goods** The removal or substitution of any **materials and goods** which do not conform to the specification or the **contract drawings** shall not constitute grounds for the extension of the **construction period** nor for the adjustment of the **contract value** [17.1.8; 23.1 & 2] F:..... V:..... T:.....		Item		**Not Priced**
24	Clause 24.0 - **Penalty** for late or non-completion F:..... V:..... T:.....		Item		**Not Priced**
	<u>Payment (A25 - A27)</u>				
25	Clause 25.0 - Payment **Prices submitted** Where prices are submitted by the **contractor** or **subcontractor** during the progress of the **works** in respect of **contract instructions** or in regard to a claim under the terms of this **agreement** and notwithstanding the fact that such prices may be used in an interim **payment certificate**, there is to be no presumption of acceptance. Should the **principal agent** wish to accept any such prices prior to the issue of the **certificate of final completion**, it shall be in writing F:..... V:..... T:.....		Item		**Not Priced**
26	Clause 26.0 - Adjustment of the **contract value** and **final account**				
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
	Fluctuations in costs All fluctuations in costs, with the exception of fluctuations in the rate of Value Added Tax, shall be for the account of the **contractor** [26.9.5]				
	Cost of claims All costs incurred by the **contractor** in the preparation of claims shall be borne by the **contractor**. This provision shall not preclude an adjudicator or an arbitrator appointed in terms of this **agreement** [30.6 & 7] from making a determination on costs				
	Claims from subcontractors The **contractor** shall review, assess and adjudicate any claims received by him from any **subcontractor** and thereafter submit same to the **principal agent** with a recommendation in order to assist the **principal agent** in adjudicating the claim [26.6]				
	F:..... V:..... T:.....		Item		**Not Priced**
27	Clause 27.0 - Recovery of expense and/or loss				
	F:..... V:..... T:.....		Item		**Not Priced**
	<u>Suspension and termination (A28 - A29)</u>				
28	Clause 28.0 - Suspension by the **contractor**				
	F:..... V:..... T:.....		Item		**Not Priced**
29	Clause 29.0 - Termination				
	F:..... V:..... T:.....		Item		**Not Priced**
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
	Dispute resolution (A30)				
30	Clause 30.0 - Dispute resolution				
	F:..... V:..... T:.....		Item		**Not Priced**
31	**Agreement**				
	The required information of the **parties** and the amount of the **contract sum** shall be inserted in the **agreement** for signature of the **agreement** by the **parties**				
	F:..... V:..... T:.....		Item		**Not Priced**
32	**Contract data**				
	Tenderer's selections				
	Before submission of his tender the **contractor** is to complete the tenderer's selections in the **contract data**				
	F:..... V:..... T:.....		Item		**Not Priced**
	SECTION B: GENERAL PRELIMINARIES				
	Definitions and interpretation (B1)				
33	Clause 1.1 - Definitions				
	F:..... V:..... T:.....		Item		**Not Priced**
34	Clause 1.2 - Interpretation				
	F:..... V:..... T:.....		Item		**Not Priced**
	Documents (B2)				
35	Clause 2.1 - Checking of documents				
	F:..... V:..... T:.....		Item		**Not Priced**
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
36	Clause 2.2 - Provisional **bills of quantities** **Multiple procurement** These **bills of quantities** are in multiple procurement format ie the "wet trades" - earthworks, concrete, formwork and reinforcement, precast concrete, masonry, waterproofing and sub-surface drainage - are provisionally measured and the subsequent trades are **budgetary allowances** and/or **provisional sums** F:..... V:..... T:.....		Item		**Not Priced**
37	Clause 2.3 - Availability of **construction information** F:..... V:..... T:.....		Item		**Not Priced**
38	Clause 2.4 - Ordering of **materials and goods** F:..... V:..... T:.....		Item		**Not Priced**
	<u>Previous work and adjoining properties (B3)</u>				
39	Clause 3.1 - Previous work - dimensional accuracy F:..... V:..... T:.....		Item		**Not Priced**
40	Clause 3.2 - Previous work - **defects** F:..... V:..... T:.....		Item		**Not Priced**
41	Clause 3.3 - Inspection of adjoining properties F:..... V:..... T:.....		Item		**Not Priced**
	Carried Forward			R	2 424 228.00

	Brought Forward		R	2 424 228.00
	The site (B4)			
42	Clause 4.1 - Handover of **site** in stages			
	F:..... V:..... T:.....	Item		**Not Priced**
43	Clause 4.2 - Enclosure of the **works**	Item		**Not Priced**
44	Clause 4.3 - Geotechnical and other investigations			
	F:..... V:..... T:.....	Item		**Not Priced**
45	Clause 4.4 - Encroachments			
	F:..... V:..... T:.....	Item		**Not Priced**
46	Clause 4.5 - Existing premises occupied			
	F:..... V:..... T:.....	Item		**Not Priced**
47	Clause 4.6 - Services - known			
	F:..... V:..... T:.....	Item		**Not Priced**
	Management of contract (B5)			
48	Clause 5.1 - Management of the **works**			
	F:..... V:..... T:.....	Item		**Not Priced**
49	Clause 5.2 - Progress meetings			
	F:..... V:..... T:.....	Item		**Not Priced**
50	Clause 5.3 - Technical meetings			
	F:..... V:..... T:.....	Item		**Not Priced**
	Carried Forward		R	2 424 228.00

	Brought Forward			R	2 424 228.00
	Samples, shop drawings and manufacturer's instructions (B6)				
51	Clause 6.1 - Samples of materials				
	F:..... V:..... T:.....		Item		**Not Priced**
52	Clause 6.2 - Workmanship samples				
	F:..... V:..... T:.....		Item		**Not Priced**
53	Clause 6.3 - Shop drawings				
	F:..... V:..... T:.....		Item		**Not Priced**
54	Clause 6.4 - Compliance with manufacturer's instructions				
	F:..... V:..... T:.....		Item		**Not Priced**
	Deposits and fees (B7)				
55	Clause 7.1 - Deposits and fees				
	F:..... V:..... T:.....		Item		**Not Priced**
	Temporary services (B8)				
56	Clause 8.1 - Water				
	F:..... V:..... T:.....		Item		**Not Priced**
57	Clause 8.2 - Electricity				
	F:..... V:..... T:.....		Item		**Not Priced**
58	Clause 8.3 - Ablution and welfare facilities				
	F:..... V:..... T:.....		Item		**Not Priced**
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
59	Clause 8.4 - Communication facilities F:..... V:..... T:.....		Item		**Not Priced**
	Prime cost amounts (B9)				
60	Clause 9.1 - Responsibility for **prime cost amounts** F:..... V:..... T:.....		Item		**Not Priced**
	Attendance on subcontractors (B10)				
61	Clause 10.1 - General attendance F:..... V:..... T:.....		Item		**Not Priced**
62	Clause 10.2 - Special attendance F:..... V:..... T:.....		Item		**Not Priced**
	General (B11)				
63	Clause 11.1 - Protection of the **works** F:..... V:..... T:.....		Item		**Not Priced**
64	Clause 11.2 - Protection/isolation of existing **works** and **works** occupied in **sections** F:..... V:..... T:.....		Item		**Not Priced**
65	Clause 11.3 - Security of the **works** F:..... V:..... T:.....		Item		**Not Priced**
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
66	Clause 11.4 - Notice before covering work F:..... V:..... T:.....		Item		**Not Priced**
67	Clause 11.5 - Disturbance **Disturbance** All work is to be carried out in such a manner as to cause no unacceptable or unreasonable dust, noise, vibrations, nuisance, inconvenience, annoyance and the like to the public, others, other properties and traffic in so far as they exceed the permissible limitations set by government legislation or by the local authority. Any delays, stoppages and the like arising from or in order to comply with the above will not constitute grounds for an adjustment to the **construction period** or **contract value** whatsoever F:..... V:.....T:.....		Item		**Not Priced**
68	Clause 11.6 - Environmental disturbance **Controlling all forms of pollution** The **contractor** shall be responsible for and take all precautions in controlling by whatever means necessary all forms of pollution emanating from the **site** during the **construction period** due inter alia to noise, artificial light, wind-blown sand, dust, deposits of mud, etc The **contractor** is to ensure that all roads which border the **site** and are used by the **contractor** during the execution of the **works** are kept clean and free of any dirt or debris caused by the execution of the **works** **Environmental management plan** The **employer** has prepared an environmental management plan (EMP) (refer to Annexure D-2 for a copy of the relevant plan). The **contractor** shall price opposite this item for compliance with all the requirements of such EMP F:..... V:.....T:.....		Item		**Not Priced**
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
69	Clause 11.7 - **Works** cleaning and clearing F:..... V:..... T:.....		Item		**Not Priced**
70	Clause 11.8 - Vermin F:..... V:..... T:.....		Item		**Not Priced**
71	Clause 11.9 - Overhand work F:..... V:..... T:.....		Item		**Not Priced**
72	Clause 11.10 - Tenant installations F:..... V:..... T:.....		Item		**Not Priced**
73	Clause 11.11 - Advertising F:..... V:..... T:.....		Item		**Not Priced**
	SECTION C: SPECIFIC PRELIMINARIES				
74	Warranties for materials and workmanship Where warranties for materials and/or workmanship are called for, the **contractor** shall obtain a written warranty, addressed to the **employer**, from the entity supplying the materials and/or executing the work and shall deliver same to the **principal agent** on **final completion** of the contract The warranty shall state that workmanship, materials and installation are warranted for a specific period from the date of **practical completion** and that any **defects** that may arise during the specified period shall be made good at the expense of the entity supplying the materials and/or doing the work, upon written **notice** to do so				
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
	The warranty will not be enforced if the work is damaged by **defects** in the execution of the **works**, in which case the responsibility for replacement shall rest entirely with the **contractor** F:..... V:..... T:.....		Item		**Not Priced**
75	Overtime Should overtime be required to be worked for any reason whatsoever, the cost of such overtime is to be borne by the **contractor** unless the **principal agent** has specifically authorised, prior to execution thereof, that costs for such overtime are to be borne by the **employer** F:..... V:.....T:.....		Item		**Not Priced**
76	Cooperation of the **contractor** for cost management It is specifically agreed that the **contractor** accepts the obligation of assisting the **principal agent** in implementing proper cost management. The **contractor** will be advised by the **principal agent** of all cost management procedures which will be implemented to ensure that the **contract value** does not exceed the budget F:..... V:.....T:.....		Item		**Not Priced**
77	Overloading The **contractor** shall take all necessary steps to ensure that no damage occurs due to overloading of any portion of the **works** or temporary works eg scaffolding, etc. The **contractor** shall submit details of his proposed loading, storage, plant erection, etc to the **principal agent** for approval prior to proceeding with such loading, storing or erecting and shall comply with and pay for the **principal agent's** requirements in connection with the provision of temporary support work, etc. Any damage caused to the **works** by overloading shall be made good by the **contractor** at his sole expense F:..... V:.....T:.....		Item		**Not Priced**
	Carried Forward			R	2 424 228.00

	Brought Forward			R	2 424 228.00
78	Propping of floors below The **contractor** is advised that propping of floors below may be required if he wishes to use any areas of completed suspended reinforced concrete slabs for vehicle access, storage of **materials and goods** and location of plant, scaffolding, etc. The location of these areas and any necessary propping shall be approved by the **principal agent** and the cost thereof shall be borne by the **contractor** F:..... V:.....T:.....		Item		**Not Priced**
79	Testing of flat roof waterproofing for watertightness Flat roof waterproof areas shall be flooded and kept "ponded" for at least forty eight (48) hours as a test to ensure the watertightness of the waterproofing and before any further construction work is carried out above the waterproofing F:..... V:.....T:.....		Item		**Not Priced**
80	Health and safety The **contractor** shall: 1. Comply with the health and safety specification for the **works** 2. Prepare and agree with the health and safety consultant the health and safety plan for the **works** 3. Cooperate with the health and safety consultant in all respects 4. Manage the compliance of all subcontractors with the regulations and with the health and safety plan and specification 5. Conform to the conditions contained in the **employer's** health and safety specification F:..... V:..... T:.....		Item		**Not Priced**
	Carried Forward			R	2 424 228.00

	Brought Forward		R	2 424 228.00
81	Green star building certification			
	F:..... V:.....T:.....	Item		**Not Priced**
82	Broad based black economic empowerment (BBBEE) Tenders submitted will be evaluated taking into account their empowerment rating The **employer** will be monitoring the broad based black economic empowerment (BBBEE) status of the **contractor** throughout the execution of the **works** The **contractor** is to submit to the **principal agent** on an annual basis a schedule of spend, split into vendors engaged as **subcontractors** and suppliers indicating their BBBEE rating including proof of the said rating			
	F:..... V:.....T:.....	Item		**Not Priced**
83	Advertising rights The **employer** may elect to contract with advertising agencies for the erection of advertising hoardings, banners, wraps or the like for the duration of the contract. The **contractor** shall not prevent such an arrangement and will assist in the facilitation of same. The position and type of advertising structure to be agreed with the **principal agent** so as not to hinder the **contractor** in meeting his obligations under this **agreement**			
	F:..... V:.....T:.....	Item		**Not Priced**
84	Confidentiality The **contractor** undertakes to maintain in confidence any and all information regarding this project and shall obtain appropriate similar undertakings from all **subcontractors** and suppliers. Such information shall not be used in any way except in connection with the execution of the **works** No information regarding this project shall be published or disclosed without the prior written consent of the **employer**			
	F:..... V:.....T:.....	Item		**Not Priced**
	Carried Forward		R	2 424 228.00

	Brought Forward			R	2 424 228.00
85	Media releases All rights of publication of articles in the media, together with any advertising relating thereto or in any way connected with this project, shall vest with the **employer** The **contractor** together with his **subcontractors** shall not, without the prior written consent of the **employer**, cause any statement or advertisement connected with this project to be printed, screened or aired by the media F:..... V:.....T:..... Allow provision for signage on the contractor's crane for the developers logo		Item		**Not Priced**
86	F:..... V:..... T:.....		Item		**Not Priced**
	SUMMARY OF CATEGORIES				
87	Category : Fixed R..... Category : Value R..... Category : Time R.....		Item		**Not Priced**
	Carried to Final Summary			R	2 424 228.00

Item No		Unit	Quantity	Rate	Amount

SECTION 2

BILL NO.1

DEMOLITIONS AND ALTERATIONS

Preambles

For preambles see "Model Preambles for Trades"

SUPPLEMENTARY PREAMBLES

View site

Before submitting his tender the contractor shall visit the site and satisfy himself as to the nature and extent of the work to be done and the value of the materials contained in the buildings or portions of the buildings to be demolished. No claim for any variations of the contract sum in respect of the nature and extent of the work or of inferior or damaged materials will be entertained

Explosives

No explosives whatsoever may be used for demolition purposes unless otherwise stated

General

The contractor shall carry out the whole of the works with as little mess and noise as possible and with a minimum of disturbance to adjoining premises and their tenants. He shall provide proper protection and provide, erect and remove when directed, any temporary tarpaulins that may be necessary during the progress of the works, all to the satisfaction of the principal agent.

Water supply pipes and other piping that may be encountered and found necessary to disconnect or cut, shall be effectually stopped off or grubbed up and removed, and any new connections that may be necessary shall be made with proper fittings, to the

Carried Forward R **Not Priced**

	Brought Forward		R	**Not Priced**
	satisfaction of the principal agent. **Doors, fanlights, fittings, frames, linings, etc which are to be re-used shall be thoroughly overhauled before re-fixing including taking off, easing and rehanging, cramping up, re-wedging as required and making good cramps, dowels, etc, and easing, oiling, adjusting and repairing ironmongery as necessary, replacing any glass damaged in removal or subsequently and stopping up all nail and screw holes with tinted plastic wood to match timber, unless otherwise described. Re-painting or re-varnishing is given separately** **Prices for taking out of doors, windows, etc shall include for removal of all beads, architraves, ironmongery, etc** **Prices for taking out and removing doors and frames** **shall include for removing door stops, cabin hooks, etc and making good floor and wall finishes to match** **existing.** **With regard to building up of openings in existing walls, cement screeds and pavings, granolithic, tops of walls, etc, shall be levelled and prepared for raising of brickwork** **Making good of finishes shall include making good of** **the brick and concrete surfaces onto which the new finishes are applied, where necessary** **The contractor will be required to take all dimensions** **affecting the existing buildings on the site and he will be held solely responsible for the accuracy of all such** **dimensions where used in the manufacture of new items (doors, windows, fittings, etc)**			
	REMOVAL OF EXISTING WORK			
	Carried Forward		R	**Not Priced**

				R	Not Priced
	Brought Forward				
	Taking down and removing roofs, floors, panelling, ceilings, partitions, etc				
1	Gypsum plasterboard ceilings, including cornices, timber brandering, etc	m2	3 164	85.00	268 940.00
	Taking out and removing sundry joinery work, fittings, etc				
2	Timber skirtings	m	691	56.00	38 696.00
	Taking out and removing sundry metal work				
3	Steel balustrades 1,8mm high from concrete stairs, including making good granolithic finish	m	670	255.00	170 850.00
	Taking out/off and removing glass and mirrors				
4	Carefully remove cracked or broken glazing in panes not exceeding 0,1m2 in existing metal windows and doors including scraping out putty on reveals and prepare windows and or doors to receive new glazing (New glazing elsewhere measured)	No	198	35.00	6 930.00
	PREPARATORY WORK TO EXISTING SURFACES				
	Making good defects in existing screeded floors with "15mm"				
5	Making good defects in existing screeded floors with "15mm"	m2	3 163	75.00	237 225.00
	CLEANING AND SERVICING OF EXISTING WORK, ETC				
	MAKING GOOD OF FINISHES, ETC				
	Taking down and removing roofs, e.t.c				
6	Carefully remove broken or damaged fibre fascia/bargeboards not exceeding 300mm wide, replace with new and paint to match existng	m	853	120.00	102 360.00
	Carried Forward			R	825 001.00

	Brought Forward			R	825 001.00
7	Carefully cut out and remove damaged sections of existing galvanised 100 x 75mm downpipes in lengths not exceeding 0,5m and replace with new to match existing profile, make water-tight at joining, prepare and paint to match existing	m	990	120.00	118 800.00
8	Carefully cut out and remove damaged sections of existing galvanised 125 x 100mm gutter in lengths not exceeding 0,5m and replace with new to match existing profile, make water-tight at joining, prepare and paint to match existing	m	853	105.00	89 565.00
9	Repair leaks and make water-tight IBR roofing sheets and paint to match existing colour	m2	1 264	180.00	227 520.00
	SERVICING EXISTING ITEMS				
	INTERNAL PLASTER				
10	Carefully hack out and remove damp, loose, flaking or cracked plaster to walls exceeding 1,0m2 and prepare surface of wall to receive new plaster (New plaster elsewhere measured)	m	720	76.00	54 720.00
	Damp proof treatment				
11	Hack-off plaster, apply damp repellant and replaster and paint to match existing	m2	1 350	218.00	294 300.00
	Carried Forward to Summary of Section 2			R	1 609 906.00

Section No. 2
Bill No.1 Demolitions and Alterations (Provisional)
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

Item No		Unit	Quantity	Rate	Amount

SECTION 2

BILL NO.2

CONCRETE, FORMWORK AND REINFORCEMENT

Preambles

For preambles see "Model Preambles for Trades"

STANDARD PREAMBLES

Note:

SUPPLEMENTARY PREAMBLES

Cost of tests

The costs of making, storing and testing of concrete test cubes as required under clause 7 "Tests" of SABS 1200 G shall include the cost of providing cube moulds necessary for the purpose, for testing costs and for submitting reports on the tests to the architect. The testing shall be undertaken by an independent firm or institution nominated by the contractor to the approval of the architect. (Test cubes are measured separately).

Formwork

Descriptions of formwork shall be deemed to include use and waste only (except where described as "left in" or "permanent"), for fitting together in the required forms, wedging, plumbing and fixing to true angles and surfaces as necessary to ensure easy release during stripping and for reconditioning as necessary before re-use.

The vertical strutting shall be carried down to such construction as is sufficiently strong to afford the required support without damage and shall remain in position until the newly constructed work is able to support itself.

Carried Forward R **Not Priced**

				R	Not Priced
	Brought Forward				
	Formwork to soffits of solid slabs etc shall be deemed to be to slabs not exceeding 250mm thick unless otherwise described Formwork to sides of bases, pile caps, ground beams, etc will only be measured where it is prescribed by the engineer for design reasons. Formwork necessitated by irregularity or collapse of excavated faces will not be measured and the cost thereof shall be deemed to be included in the allowance for taking the risk of collapse of the sides of the excavations, provision for which is made in "Earthworks"				
	REINFORCED CONCRETE CAST ON/IN FORMWORK ETC				
	30MPa/19mm concrete				
1	Stairs including landings, beams and inverted beams	m3	126	2 500.00	315 000.00
	TEST BLOCKS				
2	Making and testing 150 x 150 x 150mm concrete strength test cube	No	15	500.00	7 500.00
	CONCRETE SUNDRIES				
	Finishing top surfaces of concrete smooth with a power float				
3	Surface beds, slabs, etc to falls	m2	35	95.00	3 325.00
	ROUGH FORMWORK				
	Rough formwork to soffits				
4	Slabs propped up exceeding 1.5m and not exceeding 3.5m high	m2	90	500.00	45 000.00
	Carried Forward			R	370 825.00

	Brought Forward			R	370 825.00
	Rough formwork to sides				
5	Edges, risers, ends and reveals not exceeding 300mm high or wide (Provisional)	m	39	50.00	1 950.00
	MOVEMENT JOINTS, ETC				
	Expansion joints with bitumen impregnated softboard between vertical concrete and brick surfaces				
6	10mm Joints not exceeding 300mm high	m	95	30.00	2 850.00
	Saw cut joints				
7	3 x 40mm Saw cut joints in top of concrete	m	185	30.00	5 550.00
	REINFORCEMENT (PROVISIONAL)				
	Steel reinforcement to structural concrete work (including spacers)				
8	Various Diameter bars	t	22.00	13 500.00	297 000.00
	Carried Forward to Summary of Section 2			R	678 175.00

Item No		Unit	Quantity	Rate	Amount
	SECTION 2				
	BILL NO. 3				
	MASONRY				
	Preambles				
	For preambles see "Model Preambles for Trades"				
	SUPPLEMENTARY PREAMBLES				
	Note:				
	Sizes in descriptions				
	Where sizes in descriptions are given in brick units, "one brick" shall represent the length and "half brick" the width of a brick				
	Re-measurement of work				
	Final remeasurement of quantities to be done on a recurring basis after completion of the works and certain items (ex budgetary allowances) may be subject to a proven cost claim.				
	Note:				
	SUPERSTRUCTURE				
	ORDINARY NON-LOAD BEARING BRICKWORK				
	Brickwork				
1	Half brick wall as per specification ref (Firewall)	m2	275	345.00	94 875.00
	BRICKWORK SUNDRIES				
	Brick reinforcement and hoop iron ties				
2	75mm Wide reinforcement built in horizontally	m	716	8.00	5 728.00
	Carried Forward to Summary of Section 2			R	100 603.00

Item No		Unit	Quantity	Rate	Amount
	SECTION NO. 2				
	BILL NO. 4				
	ROOF COVERINGS, ETC:				
	Preambles				
	For preambles see "Model Preambles for Trades"				
	SUPPLEMENTARY PREAMBLES				
	Note:				
	The Contractor is to submit to the consultant Architect a certificate signed by the Merchant, stating that the galvanized roof covering supplied complies with the required thickness specified				
	All holes through profiled metal sheeting are to be drilled				
	PROFILED METAL SHEETING AND ACCESSORIES				
	Global Roofing Solutions 700mm cover Klip-Tite™ profile roll-formed in continuous lengths from 0.58mm thick Z275 spelter ISQ550 Galvanised steel roof sheeting, fixed to timber intermediate purlins at MAX 1200mm centres and eaves and ridge purlins at MAX 1200mm centres (final spacing to be calculated by an Engineer) using KL700 plus clips fixed with 10No.11 x 45mm long self-drilling wafer head PH2 screws, type 17 drill point fasteners (fastener lengths may vary depending on the insulation used or requirements), all following the manufacturer's specifications by a GRS approved contractor.				
1	Roof covering with pitches not excedding 25 degrees	m2	1 461	430.00	628 230.00
2	Extra over ridge capping	m	132	126.00	16 632.00
3	Extra over hip capping	m	15	126.00	1 890.00
	ROOF AND WALL INSULATION				
	Carried Forward			R	646 752.00

	Brought Forward			R	646 752.00
	Sisalation FR405 (Fire Retardant) aluminium foil insulation				
4	Insulation laid taut over purlins and fixed concurrent with roof covering; overlapped longitudinally by 100mm; on and including white PVC coated straining wire spaced at 383mm centres: all In strict accordance with the manufacturers specifications.	m2	1 461	150.00	219 150.00
	Carried Forward to Summary of Section 2			R	865 902.00

Section No. 2
Bill No.4 Roof Covering
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

Item No		Unit	Quantity	Rate	Amount
	SECTION 2				
	BILL NO.5				
	CARPENTRY AND JOINERY				
	Preambles				
	For preambles see "Model Preambles for Trades"				
	SUPPLEMENTARY PREAMBLES				
	Note:				
	Fascias, etc.				
	Descriptions of fibre-cement fascias and barge boards shall be deemed to include for mitres, end junctions, screws, bolts, wall anchors, holes, etc.				
	ROOFS ETC				
	Plate nailed timber roof truss construction				
1	Truss construction to double pitched roof with hipped or gable ends and 2100mm high extreme including wall plates, trusses, jack rafters, permanent bracing including 38 x 52mm purlins/battens at not exceeding 900mm centres for corrugated roofing sheets	m2	290	395.00	114 550.00
	FASCIAS AND BARGE BOARDS:				
	"Everite" medium density plain nutec-cement				
2	15 x 225mm Fascia or barge board countersunk screwed to support and Steel roof members (elsewhere measured) with one brass screw at maximum 750mm centres and jointed with and including standard aluminium half-round cover strips at all joints.	m	853	145.00	123 685.00
	DOORS, ETC				
	Carried Forward			R	238 235.00

	Brought Forward			R	238 235.00
	Solid Meranti door, Framed, Ledged and Braced with V-joint boarding, flush back				
3	Door overall size 813 x 2 032mm high	No	200	2 900.00	580 000.00
	Approved semi-solid core flush single door				
4	Toilet Door overall size 813 x 2 032mm high	No	198	1 200.00	237 600.00
	Carried Forward to Summary of Section 2			R	1 055 835.00

Item No			Quantity	Rate	Amount
	SECTION 2				
	BILL NO.6				
	CEILINGS AND PARTITIONING				
	Preambles				
	For preambles see "Model Preambles for Trades"				
	SUPPLEMENTARY PREAMBLES				
	Items, materials or methods to be used specified by trade names or catalogue numbers are only an indication of the quality required. Items, materials or methods of similar quality may be used with prior approval from the architect				
	Note:				
	Descriptions:				
	Items described as "nailed" shall be deemed to be fixed with hardened steel nails or pins or shot pinned to brickwork or concrete Items described as "plugged" shall be deemed to include screwing to fibre, plastic or metal plugs at not exceeding 600mm centres, and where described as "bolted" the bolts have been given elsewhere				
	CEILINGS, ETC:				
	NAILED UP CEILINGS:				
	6,4mm "Rhino" gypsum plasterboard with H-type pressed steel jointing strips				
1	Ceilings including 38 x 38mm sawn softwood brandering at 400mm centres in one direction in eaves soffits.	m2	3 164	195.00	616 980.00
	Carried Forward			R	616 980.00

	Brought Forward			R	616 980.00
2	Extra over ceiling for 650 x 650mm trap door of 38 x 50mm wrought softwood rebated framing with one 38 x 50mm sawn softwood cross brander covered with ceiling board and fitted flush in opening	No	15	750.00	11 250.00
	Rhino" gypsum plasterboard cornices				
	2x12mm plaster board on either side of 63.5mm aluminium drywall with steel studs inserted at 600mm centres into 104mm "Ultrasteel Drywall" steel track at top and bottom clad on both sides as described, including addition channel onal studs as necessary at abutments, ends, etc.				
3	75mm Coved cornices	m	691	85.00	58 735.00
	Carried Forward to Summary of Section 2			R	686 965.00

Section No. 2
Bill No.6 Ceilings and Partitioning
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

Item No		Unit	Quantity	Rate	Amount
	SECTION 2				
	BILL NO. 8				
	PLASTERING				
	Preambles				
	For preambles see "Model Preambles for Trades"				
	SUPPLEMENTARY PREAMBLES				
	Note:				
	Items, materials or methods to be used specified by trade names or catalogue numbers are only an indication of the quality required. Items, materials or methods of similar quality may be used with prior approval from the architect				
	Preparatory work for textured finishes:				
	Textured finishes to be applied to plaster (plaster elsewhere) All surfaces are to be inspected and approved by the Representative/Agent before any work is commenced with				
	Prices for textured finishes:				
	Prices are to include for all priming, preparatory work, etc.				
	PVC acrylic waterbased and PVC resin bonded textured finish:				
	All specialist plaster wall finishes are to be applied to the satisfaction of the Representative/Agent and executed by an approved firm of specialists, all strictly in accordance with the materials supplied and methods employed by the Manufacturer				
	INTERNAL PLASTER				
	Carried Forward			R	**Not Priced**

				R	**Not Priced**
	Brought Forward				
	Cement plaster steel trowelled, on brickwork				
1	On walls	m2	1 710	80.00	136 800.00
2	On narrow widths	m2	22	80.00	1 760.00
	EXTERNAL PLASTER				
	Cement plaster steel trowelled, on brickwork				
3	On walls	m2	602	80.00	48 160.00
4	On narrow widths including window cills and reveals	m2	56	80.00	4 480.00
	Carried Forward to Summary of Section 2			R	191 200.00

Section No. 2
Bill No.7 Plastering
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

Item No		Unit	Quantity	Rate	Amount

SECTION 2

BILL NO.8

PLUMBING FITTINGS

Preambles

For preambles see "Model Preambles for Trades"

SUPPLEMENTARY PREAMBLES

Note:

Items, materials or methods to be used specified by trade names or catalogue numbers are only an indication of the quality required. Items, materials or methods of similar quality may be used with prior approval from the architect

uPVC pipes and fittings:

Soil, waste and vent pipes and fittings shall be solvent weld jointed

Copper pipes:

Pipes shall be hard drawn and half-hard pipes of the class stated. Class 0 (thin walled hard drawn) pipes shall not be bent. Class 1 (thin walled half-hard), class 2 (half-hard) and class 3 (heavy walled half-hard) pipes shall only be bent with benders with inner and outer formers. Fittings to copper waste, vent and anti-syphon pipes, capillary solder fittings and compression fittings shall be "Cobra Watertech" type. Capillary solder fittings shall comply with ISO 2016. Only compression fittings shall be used in walls or in ground

Fixing of pipes

Unless specifically otherwise stated, descriptions of pipes shall be deemed to include fixing to walls etc, casting in, building in or suspending not exceeding 1m below suspension level

	Carried Forward			R	**Not Priced**

Brought Forward			R	**Not Priced**
Reducing fittings Where fittings have reducing ends or branches they are described as "reducing". In the case of pipes with diameters not exceeding 60mm only the largest end or branch size is given. Should the contractor wish to use other fittings and bushes or reducers he may do so on the understanding that no claim in this regard will be entertained. In the case of pipes with diameters exceeding 60mm all sizes are given and no claim for extra bushes, reducers, etc will be entertained Wire gratings Descriptions of gutter outlets etc shall be deemed to include wire balloon gratings Exposed concrete surfaces Exposed surfaces of concrete stormwater channels, cover slabs, inspection eye marker slabs, gulley tops, cleaning eye tops, catchpits, inspection chambers, etc shall be finished smooth with plaster Excavations No claim for rock excavation will be entertained unless the contractor has timeously notified the quantity surveyor thereof prior to backfilling "Soft rock" and "hard rock" shall be as defined in "Earthworks" Laying, backfilling, bedding, etc. of pipes Pipes shall be laid and bedded and trenches shall be carefully backfilled in accordance with manufacturers' instructions Where no manufacturers' instructions exist pipes shall be laid in accordance with clauses 5.1 and 5.2 of each of the following: SABS 1200 L : Medium-pressure pipelines LD : Sewers LE : Stormwater drainage Pipe trenches etc shall be backfilled in accordance with clauses 3, 5.5, 5.6, 5.7 and 7 of SABS 1200 DB : Earthworks (Pipe trenches) Pipes shall be bedded in accordance with clauses 3.1 to 3.4.1, 5.1 to 5.3 and 7 of SABS 1200 LB : Bedding (Pipes). Unless otherwise described bedding of rigid pipes shall be class B bedding				
Carried Forward			R	**Not Priced**

				R	Not Priced
	Brought Forward				
	Flush pans				
	Flush pans shall have straight or side outlets and "P" or "S" traps as necessary				
	Stainless steel basins, sinks, wash troughs, urinals, etc.				
	Units shall have standard aprons on all exposed edges and tiling keys against walls where applicable				
	Waste unions				
	Descriptions of waste unions shall be deemed to include rubber or vulcanite plugs and chains fixed to fittings				
	Note:				
	_____				
	RAINWATER DISPOSAL				
	0,6mm Galvanised sheet iron with chromadek finish				
1	125 x 125mm Galvanised gutter fixed to rafter feet	m	853	315.00	268 695.00
2	Extra on 100 x 125mm square gutter for stop ends	No	132	320.00	42 240.00
3	Extra on 125 x 125mm square gutter for oulet to 100mm diameter downpipe	No	132	320.00	42 240.00
4	100mm diameter rainwater downpipe	m	2 106	250.00	526 500.00
5	Extra on 100mm rainwater downpipe for shoe	No	132	315.00	41 580.00
6	Extra over 60mm rainwater pipe for eaves offset	No	132	315.00	41 580.00
	TRAPS, ETC.				
	Cobra" or other equally approved				
7	40 x 300mm Sink combination for double bowl with deepseal "P" trap	No	59	350.00	20 650.00
8	40 x 32mm Cobra CAT No 320 CP Bottle trap complete	No	15	800.00	12 000.00
	Carried Forward			R	995 485.00

	Brought Forward			R	995 485.00
	TAPS, VALVES, ETC				
	Brass				
9	15mm Fullway ballcock	No	59	200.00	11 800.00
	"Cobra Watertech" or other equally approved				
10	Basin taps/mixers complete set	No	59	880.00	51 920.00
11	Bath taps/mixers complete set	No	59	800.00	47 200.00
	SANITARY PLUMBING				
	PVC pipes				
12	50mm Pipes	m	165	80.00	13 200.00
13	110mm Pipes	m	15	150.00	2 250.00
	Extra over PVC pipes for fittings				
14	50mm Access Bend	No	59	95.00	5 605.00
15	110 x 50mm Reducer	No	20	55.00	1 100.00
16	110mm Pan connector	No	5	100.00	500.00
	WATER SUPPLIES				
	Extra over high density polyethylene class 16 pipes for fittings				
17	15mm Pipes chased into brick walls	m	550	22.00	12 100.00
	Extra over class "O" copper pipes for capillary fittings				
18	15mm Fittings	No	340	30.00	10 200.00
	Extra over class "0" copper pipes for brass compression fittings				
19	15mm Fittings	No	80	30.00	2 400.00
	Carried Forward			R	1 153 760.00

	Brought Forward			R	1 153 760.00
	Copper overflow and service pipes				
20	15mm Service pipe 500mm girth	No	10	250.00	2 500.00
	Sundries				
21	50mm diameter x 300mm long galvanised mild steel spout	No	396	385.00	152 460.00
	TESTING				
22	Testing water pipe system		Item		25 000.00
23	Testing sewer/waste pipe system		Item		25 000.00
	Carried Forward to Summary of Section 2			R	1 358 720.00

Item No		Unit	Quantity	Rate	Amount
	SECTION 2				
	BILL NO.9				
	PAINTWORK				
	PREAMBLES				
	For preambles see "Model Preambles for Trades"				
	SUPPLEMENTARY PREAMBLES				
	Note: Items, materials or methods to be used specified by trade names or catalogue numbers are only an indication of the quality required. Items, materials or methods of similar quality may be used with prior approval from the architect				
	Paint: All paint, complete with undercoat, primer, etc., to be used strictly in accordance with the manufacturer's specification. Only paint which bears the described trade names will be permitted on site. No substitution will be allowed **Paint:** NOTE: References to the following items have been made by trade name only, Contractors are invited to tender on other or equal approved items All paints used should have a 7 year quality guarantee				
	PAINTWORK, ETC TO PREVIOUSLY PAINTED WORK				
	ON INTERNAL FLOATED PLASTER SURFACES				
	Apply two coats of Acrylic PVA emulsion paint on work previously painted				
	Carried Forward			R	**Not Priced**

				R	Not Priced
	Brought Forward				
	Prepare surfaces and remove all loose material, apply one coat 'Plascon Merit Plaster Primer' and two coats 'Plascon Velvaglo Polyurethane Enamel' paint internally				
1	On internal walls	m2	1 522	75.00	114 150.00
2	Ditto, in narrow width not exceeding 300 mm wide	m2	150	75.00	11 250.00
	PAINT ON WOOD				
	Three coats superior quality clear gloss varnish				
3	On general surfaces of doors	m2	643	75.00	48 225.00
4	On door frames and linings	m2	109	75.00	8 175.00
	ON GYPSUM CEILINGS, COLUMNS ETC.				
	Universal under coat and two coats plascon acrylic - emulsion				
5	Ceilings, etc.	m2	1 266	75.00	94 950.00
	PAINT ON METAL				
	Spot priming defects in pre primed surfaces with zinc chromate primer and applying one undercoat and two coat high gloss enamel paint on steel				
6	On door frames	m2	109	75.00	8 175.00
	Carried Forward to Summary of Section 2			R	284 925.00

Section No. 2
Bill No.9 Paintwork
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

Item No		Unit	Quantity	Rate	Amount
	SECTION 2				
	BILL NO.10				
	GLAZING				
	PREAMBLES				
	For preambles see "Model Preambles for Trades"				
	SUPPLEMENTARY PREAMBLES				
	Note:				
	Items, materials or methods to be used specified by trade names or catalogue numbers are only an indication of the quality required. Items, materials or methods of similar quality may be used with prior approval from the architect				
	GLAZING TO METAL / WOOD WITH PUTTY OR BEADS				
	4mm Clear float glass				
1	Panes exceeding 0,1m2 and not exceeding 0,5m2	m2	178	450.00	80 100.00
	Carried Forward to Summary of Section 2			R	80 100.00

SECTION SUMMARY - BUILDING WORKS

Bill No		Page No		Amount
1	Bill No.1 Demolitions and Alterations (Provisional)	34		1 609 906.00
2	Bill No.2 Concrete, Formwork and Reinforcement	37		678 175.00
3	Bill No.3 Masonry	38		100 603.00
4	Bill No.4 Roof Covering	40		865 902.00
5	Bill No.5 Carpentry and Joinery	42		1 055 835.00
6	Bill No.6 Ceilings and Partitioning	44		686 965.00
7	Bill No.7 Plastering	46		191 200.00
8	Bill No.8 Plumbing and Drainage	51		1 358 720.00
9	Bill No.9 Paintwork	53		284 925.00
10	Bill No. 10 Glazing	54		80 100.00
	Carried to Final Summary		R	6 912 331.00

Section No. 2
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

Item No		Quantity	Rate	Amount
	SECTION 3			
	BILL NO.1			
	PROVISIONAL SUMS			
	Note:			
	1. The following budgetary allowances are for work to be designed by the Architect, executed by the Contractor and measured and valued by the Quantity Surveyor in terms of Clause 28 of the Conditions of the Contract.			
	2. All Provisional Sums are nett and do not include a 5% cash discount to the Contractor.			
	3. Under no circumstances may any Provisional Sum be extended at an amount lower than the amount stated.			
	4. Unless specific percentage mark-ups for profit and attendance are indicated in the Rate column, the amount priced by the Contractor for profit and attendance against each Provisional Sum shall be deemed to be lump sum prices and shall not be adjusted unless the scope of the sub-contract varies significantly.			
	Electrical Works			
1	Provide the sum of R 7 153 695, (Seven Million;One Hundred and One Hundred and Fifty Three Thousand;Six Hundred and Ninety Five Rand) for Electrical Works	Item		7 153 695.00
2	Profit	2.50%	7 153 695.00	178 842.38
3	Allow for general attendance	2.50%	7 153 695.00	178 842.38
	Carried Forward		R	7 511 379.76

	Brought Forward			R	7 511 379.76
	Mechanical Works				
4	Provide the sum of R 4 323 363,71 (Four Million Rand;Three Hundred and Twenty Three Thousand;Three Hundred and Sixty Three; Seventy One Cents) for Mechanical Works	Item			4 323 363.71
5	Profit	2.50%	4 323 363.71		108 084.09
6	Allow for general attendance	2.50%	4 323 363.71		108 084.09
	Structural Works				
7	Provide the sum of R 100 000,00 (One Hundred Thousand Rand) for Structural Works	Item			100 000.00
8	Profit	2.50%	100 000.00		2 500.00
9	Allow for general attendance	2.50%	100 000.00		2 500.00
	Decanting				
10	Provide the sum of R 2 500 000 (Two Million Five Hundred Thousand Rand) for Decanting	Item			2 500 000.00
11	Profit	2.50%	2 500 000.00		62 500.00
12	Allow for general attendance	2.50%	2 500 000.00		62 500.00
	Landscaping and Play Area				
13	Provide the sum of R 500 000,00 (Five Hundred Thousand Rand) for Landscaping and Playing Area	Item			500 000.00
14	Profit	2.50%	500 000.00		12 500.00
15	Allow for general attendance	2.50%	500 000.00		12 500.00
	Provisional Amounts for Community Liason Officer				
16	Provide the sum of R 90 000,00 (Ninety Thousand Rand) for CLO for a period of months , Monthly at R9000.00	Item			90 000.00
	Carried Forward			R	15 395 911.65

	Brought Forward		R	15 395 911.65
17	Profit	2.50%	90 000.00	2 250.00
18	Allow for general attendance	2.50%	90 000.00	2 250.00
	Carried to Final Summary		R	15 400 411.65

Section No. 3
Provisional Sums
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

Item No		Unit	Quantity	Rate	Amount

SECTION 4

BILL NO.1

EXTERNAL WORKS

PREAMBLES

For preambles see "Model Preambles for Trades"

SUPPLEMENTARY PREAMBLES

Items, materials or methods to be used specified by trade names or catalogue numbers are only an indication of the quality required. Items, materials or methods of similar quality may be used with prior approval from the architect

Nature of ground

a Soil investigation has been caried out, results can be obtained from the Engineers

Soil poisoning and insecticide:

All soil poisoning and insecticide to be applied under a ten year guarantee by an approved firm of Specialists

Carting away of excavated material

Descriptions of carting away of excavated material shall be deemed to include loading excavated material onto trucks directly from the excavations or, alternatively, from stock piles situated on the building site

General Notes:

1 Fill material shall be placed in lifts of not more than 150mm.
Compaction density to be controlled using DCP apparatus to a maximum of 30mm/blow
93% MOD AASHTO compaction is to be achieved.

2 Where backfill is required on both sides of walls, backfill both sides simultaneously.

	Carried Forward			R	**Not Priced**

				R	Not Priced
	Brought Forward				
	3 Where compaction of fill material behind retaining walls is done using mechanical rollers, the rollers must be kept 500mm away from the wall & this 500mm portion				
	Rock excavations				
	No claim for rock excavation will be entertained unless the contractor has timeously notified the quantity surveyor thereof prior to backfilling				
	SITE CLEARANCE				
	CLEAR SITE				
	Clear and grub site				
	Remove and grub large trees and stumps				
1	Girth over 1m up to 2m	No	1	5 000.00	5 000.00
	DISMANDLE AND REMOVE PIPELINE,CABLES				
2	Up to 300mm water pipeline	m	650	180.00	117 000.00
3	Up to 300mm sewer pipeline	m	650	50.00	32 500.00
4	Up to 50mm dia electric cable	m	140	35.00	4 900.00
	DEMOLISH AND REMOVE EXISTING STRUCTURE				
5	Existing concrete mountable kerb	m	510	65.00	33 150.00
6	Existing Stormwater Kerb inlet	No	2	35.00	70.00
7	Remove existing interlocking brick paving sidewalk, store on site and re-install to its original condition.	m2	3 200	170.00	544 000.00
8	Transport material and debris to dump site selected by contractor.		Item		200 000.00
	SUBSOIL DRAINAGE				
	Carried Forward			R	936 620.00

	Brought Forward			R	936 620.00
	Excavate in all materials for 600mm wide pipe trenches, backfill, compact and dispose of surplus/unsuitable material for perforated subsoil drainage pipes to the following depths and widths:				
9	Exceeding 0.0m but not 1.5m exceeding in compacted layers.	m3	510	85.00	43 350.00
	Extra-over items 4.1.1 for excavation in				
10	SOFT material	m3	51	200.00	10 200.00
11	INTERMEDIATE material	m3	26	350.00	9 100.00
12	HARD material	m3	18	500.00	9 000.00
	BEDDING (Pipes)				
	Provision of selected granular material for pipe bedding from the following sources:				
13	Trench excavation within 2km	m3	310	75.00	23 250.00
14	Provision of 19mm nom. size crushed stone aggregate required for bedding and backfilling of subsoil drain from commercial source.	m3	102	280.00	28 560.00
15	Supply and lay Geotextile membrane	m3	102	45.00	4 590.00
	SLOTED SUBSOIL DRAINS PIPELINES				
	GEOPIPES				
	Supply, handle, lay and bed perforated subsoil drains pipes, including couplings				
16	110mm dia uPVC	m	603	55.00	33 165.00
17	200mm dia uPVC	m	488	85.00	41 480.00
	Carried Forward to Summary of Section 4			R	1 139 315.00

Section No. 4
Bill No.1 Subsoil Drainage
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

Item No		Unit	Quantity	Rate	Amount

SECTION 4

BILL NO.2

ROADWORKS

For preambles see "Model Preambles for Trades"

SUPPLEMENTARY PREAMBLES

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Nature of ground

a Soil investigation has been caried out, results can be obtained from the Engineers

Soil poisoning and insecticide:

All soil poisoning and insecticide to be applied under a ten year guarantee by an approved firm of Specialists

Carting away of excavated material

Descriptions of carting away of excavated material shall be deemed to include loading excavated material onto trucks directly from the excavations or, alternatively, from stock piles situated on the building site

General Notes:

1 Fill material shall be placed in lifts of not more than 150mm.
Compaction density to be controlled using DCP apparatus to a maximum of 30mm/blow
93% MOD AASHTO compaction is to be achieved.

2 Where backfill is required on both sides of walls, backfill both sides simultaneously.

	Carried Forward			R	**Not Priced**

				R	Not Priced
	Brought Forward				
	3 Where compaction of fill material behind retaining walls is done using mechanical rollers, the rollers must be kept 500mm away from the wall & this 500mm portion				
	ROADS				
	Removal of topsoil to a nominal depth of 100mm, stockpile for use on sidewalks and disposal of surplus/unsuitable material				
	Road-bed preparation				
	Roadbed preparation, Scarify and recompact to 93% Mod AASHTO (100% for sand) for:				
1	150 mm thickness under roadways	m3	800	55.00	44 000.00
2	150 mm thickness under Sidewalks	m3	275	55.00	15 125.00
	EARTHWORKS				
	Cut to fill				
3	Compact to 90 % mod. AASHTO maximum density	m3	800	20.00	16 000.00
	Cut to Spoil				
4	Soft excavation	m3	2 250	95.00	213 750.00
	SUBBASE				
	Construct 150mm thick G5 subbase from commercial source and compact to 95% Mod AASHTO density				
5	100 mm thickness in sidewalks	m3	350	550.00	192 500.00
	Construct C4 subbase with G5 material from commercial sources to 97% MOD AASHTO Density				
6	150mm thickness in roads	m3	550	550.00	302 500.00
	Carried Forward			R	783 875.00

	Brought Forward			R	783 875.00
	Process material by means of:				
7	Stabilizing	m3	300	15.00	4 500.00
	Stabilizing agent:				
8	Portland cement	t	0.50	2 100.00	1 050.00
	BASE				
	Construct C2 base with G4 material from commercial sources compacted to 98% MOD AASHTO Density				
9	150 mm thickness in roads	m3	760	350.00	266 000.00
	Process material by means of:				
10	Stabilizing	m3	200	15.00	3 000.00
	Stabilizing agent:				
11	Portland cement	t	0.80	2 100.00	1 680.00
	CONCRETE KERBING AND CHANNELING				
	Supply and place 150mm wide precast SEMI-MOUNTABLE kerb with 150mm wide cast in-situ channel as detailed on drawings				
12	Straight and curves radii greater than 20m	m	1 800	140.00	252 000.00
	Supply and place precast concrete 300mm wide MOUNTABLE kerb as detailed on drawing				
13	Straight and curves radii greater than 20m	m	28	195.00	5 460.00
14	Supply and place cast in-situ concrete 200mm wide END STRIP as shown on Drawing (Class 25/20 Concrete)	m	210	170.00	35 700.00
15	Supply and place cast in-situ concrete TRANSITION kerbs 1,0m long as shown on Drawings	m	50	250.00	12 500.00
	Carried Forward			R	1 365 765.00

	Brought Forward			R	1 365 765.00
	SEGMENTED PAVING BLOCKS				
16	Construct paving complete by supplying and using 60mm thick interlocking paving block laid to constant pattern, including 20mm thick beding and jointing sand and rolling to locked up conditions plus spread and mining weed killer into the sand layer	m2	3 200	100.00	320 000.00
17	Construct paving complete by supplying and using 80mm thick interlocking paving block laid to constant pattern, including 20mm thick beding and jointing sand and rolling to locked up conditions plus spread and mining weed killer into the sand layer	m2	1 560	150.00	234 000.00
	Carried Forward to Summary of Section 4			R	1 919 765.00

Item No		Unit	Quantity	Rate	Amount

SECTION 4

BILL NO.3

STORMWATER DRAINAGE

For preambles see "Model Preambles for Trades"

SUPPLEMENTARY PREAMBLES

Items, materials or methods to be used specified by trade names or catalogue numbers are only an indication of the quality required. Items, materials or methods of similar quality may be used with prior approval from the architect

Nature of ground

a Soil investigation has been caried out, results can be obtained from the Engineers

Soil poisoning and insecticide:

All soil poisoning and insecticide to be applied under a ten year guarantee by an approved firm of Specialists

Carting away of excavated material

Descriptions of carting away of excavated material shall be deemed to include loading excavated material onto trucks directly from the excavations or, alternatively, from stock piles situated on the building site

General Notes:

1 Fill material shall be placed in lifts of not more than 150mm.
Compaction density to be controlled using DCP apparatus to a maximum of 30mm/blow
93% MOD AASHTO compaction is to be achieved.

2 Where backfill is required on both sides of walls, backfill both sides simultaneously.

Carried Forward R **Not Priced**

				R	Not Priced
	Brought Forward				
	3 Where compaction of fill material behind retaining walls is done using mechanical rollers, the rollers must be kept 500mm away from the wall & this 500mm portion				
	EARTHWORKS				
	STORMWATER PIPE TRENCHES				
	EXCAVATION				
	Excavate in all material backfill, compact and dispose of surplus/unsuitable material for pipes:				
	Stormwater pipe culverts				
1	1000mm trench width for single 450mmdia pipe to the following depths: Exceeding 1.0m but not not Exceeding 2.0m	m	500	145.00	72 500.00
2	Extra-over Items 3.3.1 to 3.3.2 for hand excavation, backfill and compaction of pipe trenches where damage may occur, including disposal of surplus material	m3	50	115.00	5 750.00
3	Extra-over Items 3.3.1 to 3.3.3 for excavation in hard material	m3	5	105.00	525.00
4	Soilcrete backfill (7%cement) at crossings with existing streets and compact to 93% Mod. AASHTO density	m3	495	170.00	84 150.00
	BEDDING (Pipes)				
	Provision of selected fill blanket to 100mm below the pipe from the following sources:				
5	Trench excavation within 2km	m3	195	85.00	16 575.00
6	Imported from commercial source	m3	239	130.00	31 070.00
	STORMWATER DRAINAGE				
	Stormwater pipes				
	Supply, handle, lay and bed concrete pipes Class 100D with ogee joints on 100mm thick bedding:				
	Carried Forward			R	210 570.00

	Brought Forward			R	210 570.00
	450mm dia				
7	Exceeding 0.5m but Not 2m exceeding	m	450	510.00	229 500.00
	Supply, install and construct stormwater inlet structures at pipe culverts complete as shown on drawings for the following depth ranges:				
8	Exceeding 0.5m but Not 2m exceeding	No	5	4 500.00	22 500.00
	STORMWATER SIDE DRAINS				
	Construct 150mm thick V- drains complete including all works, trimming, shaping and compaction to specified densities for the following (including concrete linings and mesh 617)(LABOUR INTENSIVE)				
9	1.2m wide Side drain	m	800	550.00	440 000.00
	Carried Forward to Summary of Section 4			R	902 570.00

Section No. 4
Bill No. 3 Stormwater Drainage
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

Item No		Unit	Quantity	Rate	Amount

SECTION 4

BILL NO.4

WATER RETICULATION

For preambles see "Model Preambles for Trades"

SUPPLEMENTARY PREAMBLES

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Nature of ground

a Soil investigation has been caried out, results can be obtained from the Engineers

Soil poisoning and insecticide:

All soil poisoning and insecticide to be applied under a ten year guarantee by an approved firm of Specialists

Carting away of excavated material

Descriptions of carting away of excavated material shall be deemed to include loading excavated material onto trucks directly from the excavations or, alternatively, from stock piles situated on the building site

General Notes:

1 Fill material shall be placed in lifts of not more than 150mm.
Compaction density to be controlled using DCP apparatus to a maximum of 30mm/blow
93% MOD AASHTO compaction is to be achieved.

2 Where backfill is required on both sides of walls, backfill both sides simultaneously.

	Carried Forward			R	**Not Priced**

	Brought Forward			R	**Not Priced**
	3 Where compaction of fill material behind retaining walls is done using mechanical rollers, the rollers must be kept 500mm away from the wall & this 500mm portion				
	SPECIALS AND FITTINGS				
	GATE VALVES				
	Supply and install Socket-ended Gate valves Class PN-16 anti clock-wise closing, non-rising spindle with cap top according to SABS 665				
1	110mm dia	No	13	1 850.00	24 050.00
	FIRE HYDRANTS				
2	Supply and install Fire hydrant	No	13	5 000.00	65 000.00
	Carried Forward to Summary of Section 4			R	89 050.00

Item No		Unit	Quantity	Rate	Amount

SECTION 4

BILL NO.5

SEWER RETICULATIONS

For preambles see "Model Preambles for Trades"

SUPPLEMENTARY PREAMBLES

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Nature of ground

a Soil investigation has been caried out, results can be obtained from the Engineers

Soil poisoning and insecticide:

All soil poisoning and insecticide to be applied under a ten year guarantee by an approved firm of Specialists

Carting away of excavated material

Descriptions of carting away of excavated material shall be deemed to include loading excavated material onto trucks directly from the excavations or, alternatively, from stock piles situated on the building site

General Notes:

1 Fill material shall be placed in lifts of not more than 150mm.
Compaction density to be controlled using DCP apparatus to a maximum of 30mm/blow
93% MOD AASHTO compaction is to be achieved.

2 Where backfill is required on both sides of walls, backfill both sides simultaneously.

	Carried Forward			R	**Not Priced**

				R	Not Priced
	Brought Forward				
	3 Where compaction of fill material behind retaining walls is done using mechanical rollers, the rollers must be kept 500mm away from the wall & this 500mm portion				
	DEMOLITIONS				
1	Dismantle and remove pipelines (not encased in concrete), electricity transmission lines, cables, etc.	m	120	65.00	7 800.00
2	Remove topsoil to nominal depth of 150mm and stockpile on site for re-use.	m	12	95.00	1 140.00
3	Remove existing interlocking brick paving sidewalk, store on site and re-install to its original condition.	m	12	85.00	1 020.00
	EXCAVATION				
	Excavate in all material, backfill, compact and dispose of surplus/unsuitable material; for SEWER PIPES in 750mm trench width to the following depths: (all backfilling to be undertaken with local labour)				
	EARTHWORKS (PIPE TRENCHES)				
	Excavate in all materials for trenches backfill, compact to 93% Mod AASHTO between and over pipes, and dispose of surplus/unsuitable material, for pipes:				
4	Exceeding 0.0m but not 1.5m exceeding	m	195	75.00	14 625.00
	Extra-over items for excavation in:				
5	INTERMEDIATE material	m3	15	350.00	5 250.00
	Shoring and safe guarding of trench to the satisfaction of the Engineer				
6	Soilcrete backfill (7% cement) at crossings with existing streets and compact to 93% Mod. AASHTO density where ordered by the Engineer	m3	210	130.00	27 300.00
	BEDDING (Pipes)				
7	Imported from commercial source	m3	605	160.00	96 800.00
	Carried Forward			R	153 935.00

	Brought Forward			R	153 935.00
	Provision of selected fill blanket to 300mm above the bedding from the following sources				
	Trench excavation within 2km				
8	Imported from commercial source	m3	120	160.00	19 200.00
9	Provison of 19mm nom. size crushed stone aggregate required for bedding from commercial source when instructed by the Engineer	m3	120	210.00	25 200.00
10	Encasing of pipes in Class 15/20 concrete to 150mm on all sides of pipe including excavation, formwork, flexible joints at 4m centres, etc.	m3	165	1 400.00	231 000.00
	SEWERS				
	PIPEWORK				
	Supply, lay, joint and test core flow sewer pipes bedding, in diameters:				
	(Price includes cutting, etc.)				
11	200mm dia	m	588	250.00	147 000.00
	MANHOLES				
	Precast Concrete Manholes (up to 1250mmØ) with light duty cover and frame, complete with channels, benching, etc. as per Joburg Water standard details. Rate shall include for excavation, shoring, backfilling, and compaction to 93% Mod AASHTO minimum density.				
12	1,5 m 2,0 m	No	8	850.00	6 800.00
13	2,0 m 2,5 m	No	2	1 200.00	2 400.00
	RODDING-EYE				
14	Construct special sewer manhole as detailed and specified on drawings including all connecting to main sewer for depths not exceeding 3,1m on the following pipes:	No	6	5 000.00	30 000.00
	Carried Forward			R	615 535.00

	Brought Forward		R	615 535.00
15	Make provision for testing water tightness of manholes including all work and materials necessary as specified, only where instructed by the Engineer		SUM	20 000.00
	Carried Forward to Summary of Section 4		R	635 535.00

Section No. 4
Bill No 5 Sewer Reticulation
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

Item No		Unit	Quantity	Rate	Amount

SECTION 4

BILL NO.5

FENCING

For preambles see "Model Preambles for Trades"

SUPPLEMENTARY PREAMBLES

Items, materials or methods to be used specified by trade names or catalogue numbers are only an indication of the quality required. Items, materials or methods of similar quality may be used with prior approval from the architect

Nature of ground

a Soil investigation has been caried out, results can be obtained from the Engineers

Soil poisoning and insecticide:

All soil poisoning and insecticide to be applied under a ten year guarantee by an approved firm of Specialists

Carting away of excavated material

Descriptions of carting away of excavated material shall be deemed to include loading excavated material onto trucks directly from the excavations or, alternatively, from stock piles situated on the building site

General Notes:

1 Fill material shall be placed in lifts of not more than 150mm.
Compaction density to be controlled using DCP apparatus to a maximum of 30mm/blow
93% MOD AASHTO compaction is to be achieved.

2 Where backfill is required on both sides of walls, backfill both sides simultaneously.

	Carried Forward			R	**Not Priced**

				R	Not Priced
	Brought Forward				
	3 Where compaction of fill material behind retaining walls is done using mechanical rollers, the rollers must be kept 500mm away from the wall & this 500mm portion				
	DEMOLITIONS				
1	Remove existing fence	m	25	45.00	1 125.00
	FENCING AND GATES				
2	Complete installation of 4m Sliding Gate approved sliding gate including motor and electrical works	No	1	7 000.00	7 000.00
	Supply and erect new fencing material				
3	3,5mm Concrete Palisade Fence	m	2 100	350.00	735 000.00
	Carried Forward to Summary of Section 4			R	743 125.00

Section No. 4
Bill N0.6 Fencing
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

SECTION SUMMARY - EXTERNAL WORKS

Bill No		Page No		Amount
1	Bill No.1 Subsoil Drainage	61		1 139 315.00
2	Bill N0.2 Roadworks	65		1 919 765.00
3	Bill No. 3 Stormwater Drainage	68		902 570.00
4	Bill No.4 Water Reticulation	70		89 050.00
5	Bill No 5 Sewer Reticulation	74		635 535.00
6	Bill N0.6 Fencing	76		743 125.00
	Carried to Final Summary		R	5 429 360.00

Section No. 4
BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)

FINAL SUMMARY

Section No		Page No		Amount
1	PRELIMINARIES	30		2 424 228.00
2	BUILDING WORKS	55		6 912 331.00
3	PROVISIONAL SUMS	58		15 400 411.65
4	EXTERNAL WORKS	77		5 429 360.00
	Sub-total		R	30 166 330.65
	Contingencies	Item		1 000 000.00
	Total		R	31 166 330.65
	Carried to Form of Tender		R	31 166 330.65

BILLS OF PROVISIONAL QUANTITIES
JOHANNESBURG DEVELOPMENT AGENCY (JDA)